

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js/React front end, Firebase/GCP backend) for a provider-facing **SMART on FHIR** clinical decision support app that surfaces guideline-backed medication alternatives in the EHR prescribing flow, in a fast-paced early-stage environment.
- Extending a **Firebase Cloud Functions** backend that builds medication alternatives from **RxNorm** normalization, formulary and prior-authorization logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (Python, **FastAPI**, sentence-transformers, React, **Docker**), productionizing a published NLP method into a self-serve web tool with full ownership of architecture and technical decisions delegated by the PI.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and structured data-quality warnings; grew the backend suite to **40 tests** running hermetically in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack annotation platform (**React**, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Served as primary technical point of contact for the eTriage backend and ran structured knowledge transfer, onboarding a second engineer to on-call readiness in roughly six weeks via architecture walkthroughs and progressive ownership.

Projects

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant end to end on **Cloudflare Workers** (TypeScript), using **Durable Objects** and WorkflowEntrypoint so multi-step agent tasks survive process restarts and individual tool failures, with state persisted server-side in D1 (no external database).

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** NYC TLC dataset via broadcast joins, Adaptive Query Execution, skew-join handling, and shuffle-partition tuning; profiled stage-level timings in the Spark UI to localize distributed bottlenecks.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (Python, FastAPI) with typed retrieval planning, deterministic extraction with LLM fallback, and **cell-level provenance**; deployed as a live service on **Google Cloud Run** and hardened with **150+ tests**.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Go, Python, TypeScript, JavaScript, SQL, C++, Java, Bash

Web & Frontend: React, Next.js, Node.js, Express, FastAPI, REST APIs

Cloud Native: Kubernetes, Docker, Cloudflare Workers, Durable Objects, distributed systems

ML & AI: PyTorch, sentence-transformers, LLM agents, RAG, evaluation

Cloud & DevOps: AWS, GCP, Azure, CI/CD, Git, Linux

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.